

Interference Screw Fixation of Soft Tissue Grafts in Anterior Cruciate Ligament Reconstruction: Part 2

Effect of Preconditioning on Graft Tension During and After Screw Insertion

Janne T. Nurmi,^{*†‡} DVM, PhD, Pekka Kannus,^{*†§||} MD, PhD, Harri Sievänen,[§] ScD, Timo Järvelä,^{*†} MD, PhD, Markku Järvinen,^{*†} MD, PhD, and Teppo L. N. Järvinen,^{*†a} MD, PhD
From the ^{*}Medical School and the Institute of Medical Technology, University of Tampere, Tampere, Finland, the [†]Department of Surgery, Tampere University Hospital, Tampere, Finland, the [‡]Department of Clinical Veterinary Sciences, Faculty of Veterinary Medicine, University of Helsinki, Helsinki, Finland, the [§]Bone Research Group, UKK-Institute, Tampere, Finland, and ^{||}Accident and Trauma Research Center, UKK Institute, Tampere, Finland

Background: Preconditioning of tendon grafts is believed to eliminate natural viscoelasticity of the tendons and prevent knee laxity after anterior cruciate ligament (ACL) reconstruction.

Hypothesis: Preconditioned ACL grafts maintain their initially set tension.

Study Design: Randomized experimental study.

Methods: Forty-two human anterior tibialis (AT) tendon grafts were subjected to either no preconditioning (group 1), cyclic preconditioning (group 2), or isometric preconditioning (group 3). The residual graft tension was then recorded immediately after the application of an initial graft tension of 80 N and fixation into tibia with an interference screw, as well as 10 minutes later. In another experiment, the residual graft tension was recorded 1, 10, and 60 minutes after 10 AT and quadrupled hamstring tendon (HT) grafts alone (no fixation) had been subjected to isometric preconditioning (80 N).

Results: Immediately after screw insertion, the residual (AT) graft tensions were 79 ± 19 N, 100 ± 17 N, and 102 ± 15 N in groups 1 through 3, respectively. Ten minutes later, the corresponding values were 49 ± 16 N, 60 ± 11 N, and 64 ± 12 N. For the AT and HT grafts alone, the residual graft tensions were 67 ± 2 N and 67 ± 2 N, 45 ± 2 N and 46 ± 4 N, and 29 ± 3 N and 34 ± 5 N at 1, 10, and 60 minutes, respectively.

Conclusions: A steady decrease ($\sim 60\%$ within 60 minutes after initial tensioning) occurs in the initially set tension of the soft tissue ACL grafts.

Clinical Relevance: Clinically applicable preconditioning protocols cannot eliminate the intrinsic viscoelasticity from ACL soft tissue grafts, and thus, the reasonableness of preconditioning per se is questioned in ACL reconstruction.

Keywords: pretension; knee laxity; tensioning; ACL graft fixation; biomechanics

Soft tissue grafts, that is, the hamstring or other tendon grafts, have become increasingly popular as anterior cruciate

ligament (ACL) substitutes,^{15,16,19} and interference screw fixation is a commonly used device to secure these grafts into the femoral and tibial bone tunnels.^{12,19} Progressive creep or slippage of the graft fixed in a bone tunnel is one of the most common concerns associated with the use of the soft tissue grafts as ACL reconstruction material.^{7,17,32} Besides the slippage of the graft past the screw, the well-known viscoelasticity of tendon grafts may affect the

^a Address correspondence to Teppo Järvinen, MD, PhD, Department of Surgery, University of Tampere/IMT, FIN-33014 Tampere, Finland (e-mail: teppo.jarvinen@uta.fi).

stability of the knee joint postoperatively, as viscoelastic materials elongate with time if kept under a constant load.^{6,18,29,39} Because of this graft viscoelasticity, the intra-articular portion of the tendon graft may elongate and the graft tension decrease postoperatively. To eliminate the graft viscoelasticity, pretensioning and preconditioning of the soft tissue grafts before fixation have been recommended.^{1,29,36}

The previously used terminology has been somewhat confusing, and at least such terms as preloading, pretensioning, preconditioning (cyclic and isometric), initial tension, and set force have been used. For the sake of clarity, we decided to use the following terms in this article: *pretensioning*, referring to any loading of the graft that is performed before the graft is pulled into the femoral and tibial bone tunnels (usually using a Graftmaster board or alike in the operating table), and *preconditioning*, referring to the loading of the graft that is performed once the graft has been pulled to its proper place within the bone tunnels and 1 of the 2 ends of the graft (usually the femoral) has already been fixed. The term *preconditioning* is further divided into *cyclic* and *isometric* preconditioning, the former denoting a varying loading applied in multiple consecutive bouts (cycles) and the latter denoting a constant loading applied on the graft before the final fixation of the graft. Finally, we use the term *initial tensioning* (or *initially set tension*) to refer to the tension that is applied on the graft just prior to the actual fixation (screw insertion) of the graft into the second (usually tibial) bone tunnel.

Pretensioning and preconditioning are used to eliminate as much as possible of the viscoelastic graft creep, and the initial tension is used to apply the right amount of tension to the graft just before the graft is actually securely fixed within the bone tunnel. Although an apparent controversy surrounds the details of optimal graft pretensioning, preconditioning, and initial tensioning, a clear consensus prevails that an undefined, safe window exists between too loose and too tight tendon grafts.^{1,21,35,36} However, to date there is no clinical study to show any statistically significant differences with different pretensioning, preconditioning, and initial tensioning protocols in the long-term postoperative clinical (functional) outcome after ACL reconstruction,^{15,35,36,39} and the in vivo effects are still not fully described.^{30,35}

Any possible changes in the intra-articular graft length and tension after implantation are influenced by so many variables (eg, graft material and its properties, fixation method, knee flexion angle at surgery, tunnel placement and initial fixation-to-fixation length, postoperative rehabilitation, biologic response of the graft consisting of initial ischemic necrosis and subsequent remodelling, revascularization, and ligamentization) that it is difficult to precisely determine how much tension is actually retained postoperatively within the ACL graft.^{1,8,34,39} As recently pointed out by Shino et al,³⁰ only a few published articles have dealt with the residual graft tension during and after final fixation with particular devices.

Therefore, the objective of this second part of our series of biomechanical experiments was to assess the effects of

preconditioning on the graft tension during and after interference screw insertion.

MATERIALS AND METHODS

Specimens

The tibiae of both limbs were harvested from 21 human cadavers (mean age, 40 ± 11 years; range, 17-54): 7 women (mean age, 39 ± 11 years; range, 17-49) and 14 men (mean age, 41 ± 11 years; range, 23-54); and the anterior tibialis tendons of both limbs were removed from another group of 26 human cadavers (mean age, 64 ± 12 years; range, 40-86): 7 women (mean age, 68 ± 11 years; range, 53-79) and 19 men (mean age, 62 ± 12 years; range, 40-86). The tibiae and tendons were cleared of adherent muscle fibers and surrounding soft tissues, wrapped in saline-soaked gauze, and stored frozen at -20°C in sealed plastic bags. These preservation procedures have been recommended for knee specimens intended for in vitro testing protocols of the cruciate ligaments and ligament reconstructions⁵ and have been shown not to affect the mechanical properties of bones.²⁸

In addition, the hamstring (semitendinosus and gracilis) tendons of both limbs were removed from 5 human cadavers (mean age, 63 ± 9 years; range, 53-72): 3 men (mean age, 66 ± 10 years; range, 55-72) and 2 women (mean age, 58 ± 7 years; range, 53-63).

Study Groups

The 21 pairs of specimens (tibiae and anterior tibialis tendons) previously randomized into the serial-dilation or extraction drilling (part 1) were further allocated randomly into 3 preconditioning groups (within both primary study groups, ie, the dilation and extraction groups), thus providing us with 14 specimens in 3 preconditioning study groups (groups 1, 2, and 3).

In addition, to evaluate the possible contribution of intrinsic tendon stress relaxation to the residual graft tension and to compare the biomechanical behavior of the anterior tibialis and hamstring grafts, 5 pairs of anterior tibialis and hamstring tendon grafts were used to form 2 respective soft tissue graft stress relaxation groups without interference screw fixation into bone tunnel (groups AT and HT).

Peripheral Quantitative CT Measurements

A peripheral quantitative CT scanner (XCT 3000, Stratec Medizintechnik GmbH, Pforzheim, Germany) was used to determine the volumetric trabecular bone density (in milligrams per cubic centimeters) at the proximal tibia of all the human cadaver tibiae prior to the specimen preparation, according to a protocol described previously.^{27,31} Briefly, a cross-sectional image of the proximal tibia approximately 2 cm distal from the articular surface was

scanned corresponding to the actual site of the tibial bone tunnel of an ACL reconstruction.

Specimen Preparation and Pretensioning

On the day of testing, the tendons and tibiae were thawed to room temperature. All of the specimens were kept moist with physiologic saline solution during specimen preparation, fixation procedures, and biomechanical testing. In groups 1 through 3, a looped (double-stranded) anterior tibialis tendon graft with a total graft length of 8 cm was constructed according to the technique described by Charlick and Caborn.¹⁰ In short, the tendon was folded to form a graft with 2 strands, and while maintaining constant tension on both strands, the graft was sutured at the free end for 40 mm with number 2 braided polyester suture (Fiberwire, Arthrex Inc., Naples, Florida) by using the running baseball stitch. The diameter of each graft was measured at the sutured end with a graft-sizing tube (Acufex Microsurgical Inc., Mansfield, Massachusetts). Thereafter, the graft was pretensioned using the Graftmaster II System board (Acufex) for 15 minutes, starting at 20 pounds (88 N) of tension. A previously drilled bone tunnel²⁷ (≤ 8.5 mm, average 7.8 ± 0.6 mm) was enlarged to the desired final diameter of 10 mm either by successive placement of increased diameter dilators with 0.5-mm increments (half of the specimens in every preconditioning group) or by drilling with a conventional cannulated 10-mm drill bit (half of the specimens in every preconditioning group) according to the description in the first part of this study (part 1).

Preconditioning, Initial Tensioning, and Screw Insertion

The tibia was securely mounted to the mechanical testing machine (Lloyd LR 5K, J J Lloyd Instruments, Southampton, United Kingdom) by specially designed clamps in such a position that the bone tunnel was parallel to the direction of loading (Figure 1). In groups 1 through 3, the anterior tibialis tendon graft was pulled through the tibial bone tunnel so that 30 to 35 mm of the unsutured, looped portion of the graft, corresponding to the normal ACL length,⁵ protruded from the proximal opening of the bone

TABLE 1
Preconditioning—Study Groups

Group	Preconditioning
1	No preconditioning. 80-N initial graft tension before screw insertion. Anterior tibialis tendon graft fixed into tibial bone tunnel.
2	Cyclic preconditioning (25 cycles between 0 and 80 N in 100 seconds) followed by 80-N initial graft tension before screw insertion. Anterior tibialis tendon graft fixed into tibial bone tunnel.
3	Isometric preconditioning (80 N for 100 seconds) followed by 80-N initial graft tension before screw insertion. Anterior tibialis tendon graft fixed into tibial bone tunnel.

tunnel. The proximal end of the tendon graft was connected to the load cell of the mechanical testing machine by placing a steel bar through the unsutured, looped portion of the graft. The sutured, distal end of the graft was temporarily connected to the mechanical testing machine by tying the sutured free end of the graft to the mount holding the tibia. Then, mimicking the situation in which a surgeon is tensioning the graft manually (pulling the sutures by hand or using a tensiometer), the graft was preconditioned with the mechanical testing machine according to one of the following 3 protocols described in Table 1. An initial tension of 80 N was then applied to all grafts, and a bioabsorbable interference screw (Delta Tapered Bio-Interference Screw, Arthrex) ($\varnothing$ 11 mm, length 35 mm) was inserted with a screwdriver in outside-in fashion over a guide wire between the graft and the tibial bone-tunnel wall until the tip of the screw reached the proximal bone-tunnel opening. The preconditioning and the initial tension (80 N) were thus applied on the graft and the sutures tied on the mount fixing the tibia prior to the interference screw insertion. Immediately after the completion of the screw insertion, the sutures were cut to make sure that the tension relied solely on the interference screw fixation without any possible support from the sutures, and the graft tension was recorded at this moment, as well as 10 minutes afterward.

Soft Tissue Graft Stress Relaxation

In the AT group, a looped anterior tibialis tendon graft was constructed according to the technique described above for groups 1 through 3. In the HT group, a 4-strand hamstring tendon graft with a total graft length of 8 cm was constructed according to standard principles. In short, the 4 strands were sutured at the free end for 40 mm with number 2 braided polyester suture (Fiberwire, Arthrex) by using the running baseball stitch while maintaining constant tension on all 4 strands. The diameter of each graft in the AT and HT groups was measured at the sutured end with a graft-sizing tube (Acufex). Thereafter, the graft was pretensioned using the Graftmaster II System board

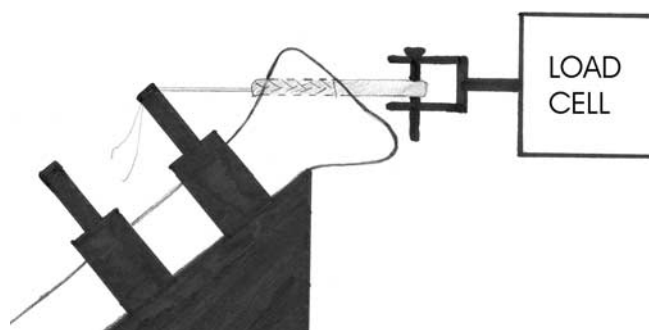


Figure 1. A schematic presentation of the biomechanical testing in groups 1 through 3.

TABLE 2
Soft Tissue Graft Stress Relaxation—Study Groups^a

Group	Preconditioning
AT	Isometric preconditioning (80 N for 100 seconds) followed by 80-N initial graft tension. Anterior tibialis tendon graft tested alone without interference screw fixation.
HT	Isometric preconditioning (80 N for 100 seconds) followed by 80-N initial graft tension. Hamstring tendon graft tested alone without interference screw fixation.

^a AT, anterior tibialis; HT, hamstring tendon.

TABLE 4
Soft Tissue Graft Stress Relaxation—Results^a

Group	Initial Graft Tension (N)	Graft Tension 1 Minute After Initial Tensioning (N) ^b	Graft Tension 10 Minutes Later (N)	Graft Tension 60 Minutes Later (N)
AT	80 N	67 ± 2.5 N	45 ± 2.2 N	29 ± 2.8 N
HT	80 N	67 ± 2.0 N	46 ± 4.1 N	34 ± 4.9N ^c

^a Mean ± SD. AT, anterior tibialis; HT, hamstring tendon.

^b Corresponding to the time required to insert the interference screw in groups 1 through 3.

^c Significantly different from group AT ($P < .02$).

(Acufex) for 15 minutes, starting at 20 pounds (88 N) of tension. To eliminate the imminent bias introduced by potential graft slippage past the interference screw, the tendon grafts in the AT and HT groups were connected directly (*no* interference screw fixation or tibial bone) to the mechanical testing machine. Briefly, a steel bar was placed through the unsutured, looped portion of the graft, and the other end of the graft (sutured portion) was mounted to the testing machine with a clamp specially designed^{22,23,37} to securely fix a soft tissue graft. A total of 30 to 35 mm of the unsutured, looped portion of the graft, corresponding to the normal ACL length,⁵ remained between the clamp and the steel bar. For the actual biomechanical testing, the grafts were held parallel to the direction of loading and preconditioned isometrically according to a protocol used for group 3 (Table 2). The residual graft tension was recorded 1 minute after 80-N initial tensioning (corresponding to the time required to insert the interference screw in groups 1-3), as well as 10 and 60 minutes thereafter.

Statistical Analysis

One-way analysis of variance with Tukey's test as the post hoc test was used to test the differences between groups 1, 2, and 3. The difference between the AT and HT groups was determined by using an unpaired sample *t* test. A *P* value of less than .05 was considered statistically significant for all tests.

TABLE 3
Preconditioning—Results^a

Group	Graft Tension Before Screw Insertion (N)	Graft Tension Immediately After Screw Insertion (N)	Graft Tension 10 Minutes After Screw Insertion (N)
1	80 N	79 ± 19 N	49 ± 16 N
2	80 N	100 ± 17 N ^b	60 ± 11 N
3	80 N	102 ± 15 N ^b	64 ± 12 N ^c

^a Mean ± SD.

^b Significantly different from group 1 ($P < .01$).

^c Significantly different from group 1 ($P < .05$).

RESULTS

Trabecular Bone Density

No significant difference was observed in the mean trabecular bone density of the region corresponding to the bone tunnel of the human cadaver tibiae, as they were 180 ± 30 mg/cm³, 182 ± 43 mg/cm³, and 176 ± 27 mg/cm³ for the 3 preconditioning groups, respectively (not significant).

Preconditioning

The graft tensions in groups 1 through 3 immediately and 10 minutes after the screw insertion are summarized in Table 3.

Soft Tissue Graft Stress Relaxation

The graft tensions in the AT and HT groups 1 minute after 80-N initial tensioning (corresponding to the time required to insert the interference screw in groups 1-3) as well as 10 and 60 minutes thereafter are summarized in Table 4.

DISCUSSION

The tendon grafts used as ACL replacement in ACL reconstruction are viscoelastic in character, and the intra-articular portion of the graft may therefore relax and elongate, resulting in a decrease in the graft tension postoperatively.^{6,18,29,39} Pretensioning and preconditioning of soft tissue grafts have been proposed as countermeasures to this undesired effect, minimizing the graft viscoelasticity-induced graft elongation so that the initially set graft tension is maintained after the final graft fixation and ultimately preventing the development of knee laxity after ACL reconstruction.^{1,29,36} In addition to preventing joint laxity, appropriate graft tension is said to be needed as stimulus for the proper orientation of the newly formed collagen during the remodelling phase of the graft.^{14,35} Conversely, an inappropriate (too high or low) initial graft tension can lead to abnormal knee kinematics and have adverse effects on the postoperative remodelling process and the mechanical properties of the graft, which, in turn, can secondarily result in increased laxity.^{1,2,21,26,34,38,39}

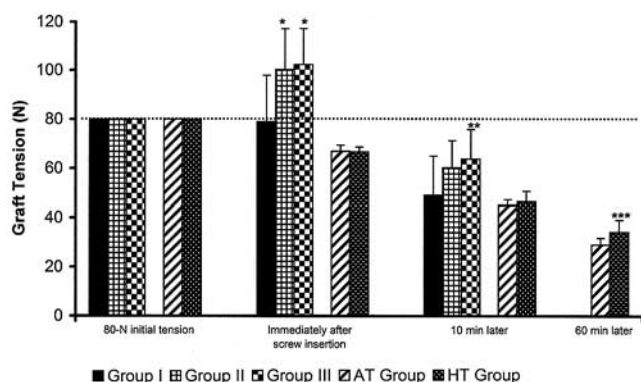


Figure 2. The average graft tensions (mean $\pm$ SD) in groups 1 through 3, AT, and HT after initial tensioning (80 N), immediately after interference screw insertion (1 minute after the initial tensioning in AT and HT), 10 minutes after interference screw insertion, and 60 minutes after initial tensioning (AT and HT only). *, Significantly different from group 1 ($P < .01$); **, Significantly different from group 1 ($P < .05$); ***, Significantly different from group AT ($P < .05$).

However, it is not known how the graft tension actually changes during graft healing and remodelling and what the optimal graft tension after ACL reconstruction should be.

In this study, we wanted to investigate the effects of different preconditioning protocols on the residual soft tissue graft tension when interference screw fixation is used as a sole means of fixation in ACL reconstruction. In an attempt to eliminate the tendon graft viscoelasticity-induced decrease in the graft tension, we used relatively intensive pretensioning and preconditioning protocols, but such that can also be used clinically using commercially available graft preparation boards and tensiometers. After pretensioning and preconditioning, the initial graft tension recommended by Yasuda et al³⁶ (80 N) was applied to all grafts before final fixation of the graft.

Preconditioning and interference screw insertion temporarily increased the residual graft tension, but the remaining graft tension was considerably decreased in all study groups irrespective of preconditioning status (whether or not preconditioned) already 10 minutes after the tensioning (Figure 2). One can naturally speculate that the observed decrease in the residual graft tension occurred as the pretensioning and preconditioning protocols used in this study were not sufficiently intensive to properly eliminate the intrinsic creep of the tendon grafts or, alternatively, as the decrease in tension was attributable to preconditioning-induced graft deterioration or slippage past the screw. Regarding the possible graft deterioration, it is highly unlikely that graft deterioration would have occurred at the above-described low load levels (<100 N) as the average ultimate failure load of anterior tibialis tendon graft has been reported to be more than 4000 N.¹³ Furthermore, the results of our experiment with HT and AT grafts only, showing that despite the identical preconditioning of that used in group 3, more than 60% of the initially set graft tension was lost within 1 hour even when

the chance of graft slippage was eliminated, suggest that the remaining intrinsic graft stress relaxation is a more plausible explanation for the dramatic decrease in graft tension observed in all study groups (1-3).

Our findings are actually in agreement with a study by Schatzmann et al,²⁹ in which the structural and mechanical properties of quadriceps tendons and patellar ligaments were compared using cyclic preconditioning with an extremely intensive loading protocol (cycling between 75 and 800 N at 0.5 Hz frequency). The authors reported that more than 160 cycles were needed to reach a steady state where no further graft elongation occurred,²⁹ a situation that is obviously impossible to reach clinically. Gradually decreasing residual tension has previously also been reported by Shino et al³⁰ who fixed, according to the terminology used in the current study, isometrically preconditioned (49 N or 98 N for 5 minutes) double-looped bovine tendons into porcine bone tunnel with a new fixation device, Double Spike Plate. In their study, graft tension was observed to increase immediately after fixation but to decrease back to the initially set graft tension level in 5 minutes.³⁰ Furthermore, regarding the mode of preconditioning, the cyclic preconditioning has been proposed to be particularly efficient in eliminating graft viscoelasticity,²⁹ but we did not find any differences between cyclic and isometric preconditioning protocols (groups 2 and 3). Similarly, Graf et al reported that simple isometric preconditioning was equally effective to cyclic stretching in a study by using primate patellar tendon graft.¹⁸ Altogether, it is most likely that the clinically applicable pretensioning, preconditioning, and initial tensioning protocols cannot fully eliminate intrinsic tendon viscoelasticity or provide any clear advantage over simple tensioning of the graft (eg, pulling by hand) in ACL reconstruction, and thus, the clinical reasonability of graft preconditioning per se is seriously questioned.

Regarding the use of Graftmaster board, one should realize that the initially set Graftmaster tension gradually decreases over time as the graft is stretched and the sutures are tightened around the tendon bundle, and accordingly, it would be misleading to claim that the graft was tensioned with a constant load during the whole pretensioning time. Tension of 15 to 20 pounds (44-88 N) for 10 to 20 minutes using Graftmaster board has been previously clinically used to pretension hamstring tendon grafts.^{3,4,25} In this study, we used Graftmaster board primarily to ensure that all grafts and their sutures were equally pretensioned before preconditioning.

The validity of our results using the anterior tibialis tendon in relation to hamstring grafts could be questioned. However, Caborn and Selby⁹ recently described the technical suitability of the single-loop anterior tibialis tendon allografts in ACL reconstruction. Also, the similarity of the structural, material, and viscoelastic properties of anterior tibialis and quadrupled hamstring tendon grafts at the time of implantation have been shown in a comprehensive biomechanical study.¹³ Finally, the anterior tibialis and hamstring tendon grafts showed a very comparable response to retaining tendon tension, although a statistically significant difference was found between the 2 graft

types in the residual graft tension 60 minutes after initial tensioning. In this respect, one should recall that our observation period was only 60 minutes, and both grafts had already lost more than 60% of their initially set tension, suggesting that a difference of this magnitude (5 N) is likely of marginal clinical relevance.

The use of relatively large (10 mm in diameter) drill holes in this study could be considered to somehow hamper the comparison of the different preconditioning protocols, especially as it is currently recommended clinically that the diameter of the bone tunnel should closely match the diameter of the graft.³³ However, as the decrease in residual graft tension was similar among all study groups (even in the AT and HT groups, that is, when the tension was determined in grafts not fixed into tibial bone tunnel), we feel that it is warranted to conclude that the apparent graft-tunnel mismatch had no influence on the results of this study.

Although an obvious controversy seems to exist concerning the ideal pretensioning, preconditioning, and initial tensioning of the soft tissue grafts in ACL reconstruction, consensus prevails that there is a relatively wide safe window between too loose and too tight grafts.^{1,21,35,36} To date, no clinical study has managed to show any statistically significant differences in the postoperative clinical (functional) outcome between different pretensioning, preconditioning, and initial tensioning protocols,^{15,35,36,39} although Yasuda et al observed in their study of autogenous doubled semitendinosus and gracilis tendons connected in series with polyester tape that patients with an 80-N initial tension showed significantly less laxity at 2 years postoperatively than patients with a 20-N or 40-N initial tension.³⁶ Based on their results, Yasuda et al recommended that 80-N initial tension should be used to obtain an excellent outcome, especially concerning stability of the knee, in ACL reconstruction with the autogenous doubled semitendinosus and gracilis tendons.³⁶ However, initial tension higher than 80 N has been shown not to provide any further advantage²⁶ but rather to cause adverse effects on the graft.^{21,36,38} In this respect, the current clinical practice seems to well follow the general guidelines proposed by the laboratory investigations, as it was recently presented at a scientific workshop of reconstruction of the cruciate ligaments that the average initial graft tension for hamstring tendon grafts used by the surgeons is 70 N.¹ Also, Cunningham et al published a recent survey evaluating graft tensioning and found that the average normal initial tension used by sports medicine-trained orthopaedists is 60 ± 29 N.¹¹ Finally, the clinical results associated with the most commonly used tensioning protocol, the unmeasured initial tension or maximal sustained one-handed pull, have been generally good.^{11,20,24,39} Considering the results of this study along with all the above noted, it appears that if only tensioned intraoperatively within the relatively wide safe window, the soft tissue grafts will adapt to the appropriate postoperative tension for ideal healing through the intrinsic stress relaxation (viscoelasticity) of the tendons.

In conclusion, clinically applicable preconditioning protocols do not fully eliminate tendon viscoelasticity. Whether the soft tissue graft is preconditioned or not, the initially set tension decreases considerably (more than –60%) postoperatively due to the remaining intrinsic tendon creep, thus seriously questioning the reasonableness of preconditioning per se in ACL reconstruction.

ACKNOWLEDGMENT

The authors want to thank Dennis Donnermeyer, Arthrex Inc, and Jani Toukosalo, Smith & Nephew Ltd., for providing the instruments for the study. This work was supported by grants from the Finnish Foundation for Orthopaedic and Traumatological Research, the Research Council for Physical Education and Sports, Ministry of Education, the Medical Research Fund of Tampere University Hospital, and the Finnish Cultural Foundation.

REFERENCES

- Amis AA, Jakob RP. Anterior cruciate ligament graft positioning, tensioning and twisting. *Knee Surg Sports Traumatol Arthrosc.* 1998;6(suppl 1):S2-S12.
- Arms SW, Pope MH, Johnson RJ, et al. Analysis of ACL failure strength and initial strains in the canine model. *Trans Orthop Res Soc.* 1990;15:524.
- Aune AK, Holm I, Risberg MA, et al. Four-strand hamstring tendon autograft compared with patellar tendon-bone autograft for anterior cruciate ligament reconstruction: a randomized study with two-year follow-up. *Am J Sports Med.* 2001;29:722-728.
- Barrett GR, Noojin FK, Hartzog CW, et al. Reconstruction of the anterior cruciate ligament in females: a comparison of hamstring versus patellar tendon autograft. *Arthroscopy.* 2002;18:46-54.
- Beynon BD, Amis AA. In vitro testing protocols for the cruciate ligaments and ligament reconstructions. *Knee Surg Sports Traumatol Arthrosc.* 1998;6(suppl 1):S70-S76.
- Beynon BD, Johnson RJ, Fleming BC, et al. The measurement of elongation of anterior cruciate-ligament grafts in vivo. *J Bone Joint Surg.* 1994;76A:520-531.
- Brand J, Weiler A, Caborn DNM, et al. Graft fixation issues in knee ligament surgery. *Oper Tech Orthop.* 1999;9:256-263.
- Burks RT, Leland R. Determination of graft tension before fixation in anterior cruciate ligament reconstruction. *Arthroscopy.* 1988;4:260-266.
- Caborn DNM, Selby JB. Allograft anterior tibialis tendon with bioabsorbable interference screw fixation in anterior cruciate ligament reconstruction. *Arthroscopy.* 2002;18:102-105.
- Charlick DA, Caborn DNM. Alternative soft-tissue graft preparation technique for cruciate ligament reconstruction. *Arthroscopy.* 2000;16:3-5.
- Cunningham R, West JR, Greis PE, et al. A survey of the tension applied to a doubled hamstring tendon graft for reconstruction of the anterior cruciate ligament. *Arthroscopy.* 2002;18:983-988.
- DeLay BS, Smolinski RJ, Wind WM, et al. Current practices and opinions in ACL reconstruction and rehabilitation: results of a survey of the American Orthopaedic Society for Sports Medicine. *Am J Knee Surg.* 2001;14:85-91.
- Donahue TLH, Howell SM, Hull ML, et al. A biomechanical evaluation of anterior and posterior tibialis tendons as suitable single-loop anterior cruciate ligament grafts. *Arthroscopy.* 2002;18:589-597.

14. Frank C, Amiel D, Akeson WH. Healing of the medial collateral ligament of the knee: a morphological and biomechanical assessment in rabbits. *Acta Orthop Scand*. 1983;54:917-923.
15. Fu FH, Bennett CH, Lattermann C, et al. Current trends in anterior cruciate ligament reconstruction. Part I: biology and biomechanics of reconstruction. *Am J Sports Med*. 1999;27:821-830.
16. Fu FH, Bennett CH, Ma CB, et al. Current trends in anterior cruciate ligament reconstruction. Part II: operative procedures and clinical correlations. *Am J Sports Med*. 2000;28:124-130.
17. Fu FH, Ma CB. Anterior cruciate ligament reconstruction using quadruple hamstring. *Oper Tech Orthop*. 1999;9:264-272.
18. Graf BK, Vanderby R Jr, Ulm MJ, et al. Effect of preconditioning on the viscoelastic response of primate patellar tendon. *Arthroscopy*. 1994;10:90-96.
19. Harner CD, Fu FH, Irrgang JJ, et al. Anterior and posterior cruciate ligament reconstruction in the new millennium: a global perspective. *Knee Surg Sports Traumatol Arthrosc*. 2001;9:330-336.
20. Howell SM, Gittins ME, Gottlieb JE, et al. The relationship between the angle of the tibial tunnel in the coronal plane and loss of flexion and anterior laxity after anterior cruciate ligament reconstruction. *Am J Sports Med*. 2001;29:567-574.
21. Katsuragi R, Yasuda K, Tsujino J, et al. The effect of nonphysiologically high initial tension on the mechanical properties of in situ frozen anterior cruciate ligament in a canine model. *Am J Sports Med*. 2000;28:47-56.
22. Kousa P, Järvinen TLN, Vihavainen M, et al. The fixation strength of six different hamstring graft fixation devices in anterior cruciate ligament reconstruction. Part I: femoral site. *Am J Sports Med*. 2003;31:174-181.
23. Kousa P, Järvinen TLN, Vihavainen M, et al. The fixation strength of six different hamstring graft fixation devices in anterior cruciate ligament reconstruction. Part II: tibial site. *Am J Sports Med*. 2003;31:182-188.
24. Nabors ED, Richmond JC, Vannah WM, et al. Anterior cruciate ligament graft tensioning in full extension. *Am J Sports Med*. 1995;23:488-492.
25. Noojin FK, Barrett GR, Hartzog CW, et al. Clinical comparison of intra-articular anterior cruciate ligament reconstruction using autogenous semitendinosus and gracilis tendons in men versus women. *Am J Sports Med*. 2000;28:783-789.
26. Numazaki H, Tohyama H, Nakano H, et al. The effect of initial graft tension in anterior cruciate ligament reconstruction on the mechanical behaviors of the femur-graft-tibia complex during cyclic loading. *Am J Sports Med*. 2002;30:800-805.
27. Nurmi JT, Kannus P, Sievänen H, et al. Compaction drilling does not increase the initial fixation strength of the hamstring graft in ACL reconstruction in a cadaver model. *Am J Sports Med*. 2003;31:353-358.
28. Pelker RR, Friedlaender G, Markham TC, et al. Effects of freezing and freeze-drying on the biomechanical properties of rat bone. *J Orthop Res*. 1984;1:405-411.
29. Schatzmann L, Brunner P, Stäubli HU. Effect of cyclic preconditioning on the tensile properties of human quadriceps tendons and patellar ligaments. *Knee Surg Sports Traumatol Arthrosc*. 1998;6(suppl 1):S56-S61.
30. Shino K, Mae T, Maeda A, et al. Graft fixation with predetermined tension using a new device, the Double Spike Plate. *Arthroscopy*. 2002;18:908-911.
31. Sievänen H, Koskue V, Rauho A, et al. Peripheral quantitative computed tomography in human long bones: evaluation of in vitro and in vivo precision. *J Bone Miner Res*. 1998;13:871-882.
32. Simonian PT, Sussmann PS, Baldini TH, et al. Interference screw position and hamstring graft location for anterior cruciate ligament reconstruction. *Arthroscopy*. 1998;14:459-464.
33. Steenlage E, Brand JC Jr, Johnson DL, et al. Correlation of bone tunnel diameter with quadrupled hamstring graft fixation strength using a biodegradable interference screw. *Arthroscopy*. 2002;18:901-907.
34. Tohyama H, Yasuda K. Significance of graft tension in anterior cruciate ligament reconstruction: basic background and clinical outcome. *Knee Surg Sports Traumatol Arthrosc*. 1998;6(suppl 1):S30-S37.
35. van Kampen A, Wymenga AB, van der Heide HJL, et al. The effect of different graft tensioning in anterior cruciate ligament reconstruction: a prospective randomized study. *Arthroscopy*. 1998;14:845-850.
36. Yasuda K, Tsujino J, Tanabe Y, et al. Effects of initial graft tension on clinical outcome after anterior cruciate ligament reconstruction: autogenous doubled hamstring tendons connected in series with polyester tapes. *Am J Sports Med*. 1997;25:99-106.
37. Weiler A, Hoffmann RFG, Stähelin AC, et al. Hamstring tendon fixation using interference screws: a biomechanical study in calf tibial bone. *Arthroscopy*. 1998;14:29-37.
38. Yoshiya S, Andrich JT, Manley MT, et al. Graft tension in anterior cruciate ligament reconstruction: an in vivo study in dogs. *Am J Sports Med*. 1987;15:464-470.
39. Yoshiya S, Kurosaka M, Ouchi K, et al. Graft tension and knee stability after anterior cruciate ligament reconstruction. *Clin Orthop*. 2002;394:154-160.